

Ramakrishna Vennam

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SUMMARY

AI and Machine Learning professional with a proven track record of delivering impactful results through innovative data-driven solutions. Improved model accuracy by 15% while simultaneously reducing training time by 20%. Highly skilled in developing and implementing machine learning algorithms, with expertise in natural language processing and data visualization. Focused on leveraging data insights to drive business growth and efficiency.

SKILLS

JavaScript, TypeScript, React, Node.js, Python, Docker, Git, Agile/Scrum, AutoGen, FastAPI, Flask, MLOps, Azure DevOps, GitHub Actions, Azure Web Apps, Azure Functions, ACR, SQL, Generative AI, Machine learning, Deep learning, NLP, Data Visualization, Azure Cloud, Agents, Semantic Kernel, Azure AI Search, RAG, LLMs, Azure VMs

WORK EXPERIENCE

Data Scientist — Charles Schwab
Tampa, FL

Jan 2025–Present

- Engineered and deployed a Multi-Agent Chatbot utilizing Semantic Kernel and AutoGen, achieving 98% accuracy in automating enterprise-level data queries and reducing manual workload by 40%
- Architected Retrieval Augmented Generation (RAG) pipelines integrated with Azure AI Search and OpenAI LLMs, enhancing intelligent knowledge retrieval and contextual response generation for 5,000+ users
- Implemented and deployed Generative AI pipelines to process structured and unstructured data from databases, PDFs, and Word documents, ensuring 90% accuracy through model tuning and embedding optimization
- Integrated AI models into production environments using Docker, Azure DevOps, and Kubernetes, reducing deployment time by 50% and accelerating time-to-market for critical applications
- Constructed intelligent AutoGen Agents for financial data queries, resulting in a 30% improvement in decision-making efficiency for financial analysts and portfolio managers
- Created real-time Power BI dashboards integrating model insights for risk assessment and operational intelligence

Data Scientist — Accenture
India

May 2020–Jul 2023

- Developed and deployed machine learning models, including Decision Trees and Random Forest, optimizing marketing campaigns and increasing customer targeting precision by 10%
- Built end-to-end machine learning pipelines utilizing SQL, Pandas, and Power BI to perform behavioral segmentation, resulting in a 10% improvement in customer lifetime value (LTV)
- Implemented robust MLOps pipelines leveraging FastAPI, Docker, and Azure DevOps, reducing model deployment time by 60% and improving overall system reliability
- Engineered Natural Language Processing (NLP) pipelines for unstructured text data, incorporating Word2Vec, TF-IDF, and tokenization techniques to enhance model feature quality
- Directed a Deep Learning Optical Character Recognition (OCR) project using YOLO and cus-

tom PaddleOCR models, achieving 95% text extraction accuracy

PROJECTS

Multi-Agent Chatbot with RAG (AutoGen, Semantic Kernel, Azure AI Search, RAG, Docker, Azure Functions, FastAPI)

- Implemented end-to-end RAG pipeline for structured/unstructured data ingestion and model tuning.
- Integrated Azure AI Search and OpenAI GPT models for contextual document retrieval and summarization.
- Deployed to Azure Web App and Function App, achieving 90% precision in automated response generation.
- Implemented a comprehensive Retrieval-Augmented Generation (RAG) pipeline leveraging Azure AI Search and OpenAI GPT models for intelligent document retrieval and summarization, enabling automated response generation from structured and unstructured data sources.
- Developed an end-to-end RAG system using AutoGen and Semantic Kernel, integrating Azure AI Search for contextual document indexing and retrieval to enhance the accuracy and relevance of model responses.
- Deployed a scalable RAG application on Azure Web App and Function App using Docker containers, achieving 90% precision in automated response generation while handling up to 500 requests per minute, improving efficiency in information retrieval.

Generative AI for Knowledge Automation (Generative AI, Azure OpenAI, RAG, Data Preprocessing, Flask, Azure Functions)

- Built RESTful API aggregating data from multiple weather services
- Implemented caching layer improving response time by 80%
- Documented API using Swagger/OpenAPI specification

Deep Learning OCR & Object Detection (YOLO, PaddleOCR, Azure VM, Jetson, Computer Vision, Azure DevOps)

- Developed a document digitization pipeline leveraging YOLO and PaddleOCR models to automate text extraction, increasing document processing efficiency by 40% and reducing manual data entry errors.
- Implemented an automated deployment process via Azure DevOps, provisioning OCR models on Azure VMs (Jetson) and achieving 95% accuracy in character recognition across diverse document formats, improving data quality for downstream applications.
- Enhanced application efficiency for document pre-check-in processes by integrating the OCR pipeline, decreasing processing time by 60% and enabling faster document verification for over 100 users daily.

EDUCATION

University of South Florida

Master of Science in Computer Science

- Specialized in Machine Learning and Deep Learning, completing advanced coursework in Data Mining and Statistical Modeling.

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals, Data Science and Machine Learning Bootcamp with Python, Workshop in Machine Learning – Pantech E-Learning